WORKED SOLUTION

STEP III 1989, Question 13

Mechanics • Simple Harmonic Motion

STEP III 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q13

The key idea

Each string runs from O (under the table) up through its hole and across the table to P. The piece under the table always has length exactly a, the natural length, so the extension of each string is simply the distance from P to its hole. That makes each tension proportional to a vector, and with position vectors the five forces add up in one line, because the five holes are symmetrically placed and their position vectors sum to zero.

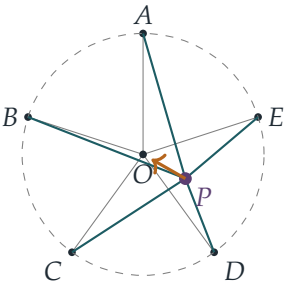
Solution

Setting up

Model P as a particle on a smooth table, the strings as light elastic strings and the holes as smooth, so the tension is the same all along each string. Take O as origin on the table, let $\mathbf{r}$ be the position vector of P and let $\mathbf{a}_1, \ldots, \mathbf{a}_5$ be the position vectors of $A, \ldots, E$, each of length a.

The string through hole i consists of a vertical piece of length a from O (under the table, directly below the centre) to the hole, and a piece on the table of length $|\mathbf{a}_i - \mathbf{r}|$. Its total length is $a + |\mathbf{a}_i - \mathbf{r}|$, so its extension is $|\mathbf{a}_i - \mathbf{r}|$. This is never negative, so every string is taut (or just slack, when P is at the hole) wherever P is: no string ever goes slack and changes the equations. By Hooke's law the tension is $\frac{\lambda}{a}|\mathbf{a}_i - \mathbf{r}|$, and on the table it pulls P towards the hole, in the direction of the unit vector $\frac{\mathbf{a}_i - \mathbf{r}}{|\mathbf{a}_i - \mathbf{r}|}$. So the force on P from string i is

$$\frac{\lambda}{a}|\mathbf{a}_i - \mathbf{r}| \cdot \frac{\mathbf{a}_i - \mathbf{r}}{|\mathbf{a}_i - \mathbf{r}|} = \frac{\lambda}{a}(\mathbf{a}_i - \mathbf{r}).$$



The teal lines are the five strings on the table, seen from above; the amber arrow shows the resultant force on P, which points straight at O.

The first part

The vertical forces on P (weight and normal reaction) balance. Adding the five horizontal forces,

$$\mathbf{F} = \frac{\lambda}{a}\sum_{i=1}^{5}(\mathbf{a}_i - \mathbf{r}) = \frac{\lambda}{a}\Big(\sum_{i=1}^{5}\mathbf{a}_i\Big) - \frac{5\lambda}{a}\mathbf{r}.$$

Ask yourself Why is $\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4 + \mathbf{a}_5 = \mathbf{0}$?

Rotating the whole figure through 72° about O permutes the five holes, so it leaves the sum $\mathbf{s} = \sum \mathbf{a}_i$ unchanged. But the only vector unchanged by a rotation through 72° is the zero vector, so $\mathbf{s} = \mathbf{0}$. (Equivalently, the fifth roots of unity sum to zero.)

Hence Newton's second law gives

$$m\ddot{\mathbf{r}} = -\frac{5\lambda}{a}\mathbf{r}, \qquad \text{that is} \qquad \ddot{\mathbf{r}} = -\omega^2\mathbf{r} \quad \text{with} \quad \omega^2 = \frac{5\lambda}{am}.$$

P starts from rest at F, so $\mathbf{r}(0) = \overrightarrow{OF}$ and $\dot{\mathbf{r}}(0) = \mathbf{0}$. The solution is $\mathbf{r} = \cos(\omega t)\,\overrightarrow{OF}$, so P moves to and fro along the straight line FO (to the reflection of F in O and back), and its displacement from O satisfies the simple harmonic equation. The period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{am}{5\lambda}},$$

as required. (Note that this works for *any* starting point F, not just for small displacements: the force law is exactly linear.)

The final part

Now string PAO has modulus $k\lambda$, so its force on P is $\frac{k\lambda}{a}(\mathbf{a}_1 - \mathbf{r})$. Using $\mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4 + \mathbf{a}_5 = -\mathbf{a}_1$,

$$\begin{aligned}\mathbf{F} &= \frac{k\lambda}{a}(\mathbf{a}_1 - \mathbf{r}) + \frac{\lambda}{a}\sum_{i=2}^{5}(\mathbf{a}_i - \mathbf{r}) \\ &= \frac{k\lambda}{a}(\mathbf{a}_1 - \mathbf{r}) + \frac{\lambda}{a}(-\mathbf{a}_1 - 4\mathbf{r}) \\ &= \frac{\lambda}{a}[(k-1)\mathbf{a}_1 - (k+4)\mathbf{r}] = -\frac{(k+4)\lambda}{a}(\mathbf{r} - \mathbf{r}_0), \qquad \mathbf{r}_0 = \frac{k-1}{k+4}\mathbf{a}_1.\end{aligned}$$

So the equilibrium position is the point $\mathbf{r}_0$ on OA (where $\mathbf{F} = \mathbf{0}$), and

$$\ddot{\mathbf{r}} = -\frac{(k+4)\lambda}{am}(\mathbf{r} - \mathbf{r}_0).$$

If P is displaced along OA from $\mathbf{r}_0$ and released from rest, then $\mathbf{r} - \mathbf{r}_0$ starts parallel to $\mathbf{a}_1$ and the acceleration is always parallel to $\mathbf{r} - \mathbf{r}_0$, with zero initial velocity. So, exactly as before, $\mathbf{r} - \mathbf{r}_0 = \cos(\omega' t)(\mathbf{r}(0) - \mathbf{r}_0)$: P moves in a straight line along OA, with simple harmonic motion about $\mathbf{r}_0$ of angular frequency ω', where

$$\omega'^2 = \frac{(k+4)\lambda}{am}.$$

The period is $T/2$ exactly when $\omega' = 2\omega$, that is, $\omega'^2 = 4\omega^2$:

$$\frac{(k+4)\lambda}{am} = \frac{20\lambda}{am} \quad \Longrightarrow \quad k + 4 = 20 \quad \Longrightarrow \quad k = 16.$$

With $k = 16$ the equilibrium position is $\mathbf{r}_0 = \frac{15}{20}\mathbf{a}_1$, three-quarters of the way from O to A. (For the model to hold, the amplitude must be less than $\frac{1}{4}a$, so that P does not reach the hole at A.)

Checks

- Setting $k = 1$ recovers the first part: $\mathbf{r}_0 = \mathbf{0}$ and $\omega'^2 = 5\lambda/(am)$.
- Units: λ is a force, so $\lambda/(am)$ has units $\mathrm{N}/(\mathrm{m\,kg}) = \mathrm{s}^{-2}$, and T is a time.
- We summed the five tension vectors at 1000 random positions (with $a = 1.3$, $m = 0.7$, $\lambda = 2.1$) and got exactly $-5\lambda\mathbf{r}/a$, and for $k = 16$ exactly $-20\lambda(\mathbf{r} - \mathbf{r}_0)/a$. Stepping the motion numerically from $(0.4, -0.3)$ returned P to its start after time $T \approx 1.85$, staying on the line through O; from $\mathbf{r}_0 + 0.1a\,\hat{\mathbf{a}}_1$ with $k = 16$ it returned after $T/2$.

Common slips

- Taking the extension to be $|\mathbf{a}_i - \mathbf{r}| - a$ or $|\mathbf{r}|$: the natural length a is used up exactly by the piece under the table.

- Resolving the five tensions with angles and trigonometry. It can be done, but vectors make the cancellation immediate.
- Assuming the new oscillation is still centred at O: the stiffer string drags the equilibrium position towards A.

Beyond STEP

A force $\mathbf{F} = -c\,\mathbf{r}$ in two dimensions gives the isotropic harmonic oscillator: the general motion (with a sideways initial velocity) is an ellipse centred at the equilibrium point, and every orbit has the same period. Together with the inverse-square law, it is one of only two central forces for which every bounded orbit closes (Bertrand's theorem). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats oscillators and central forces in this vector style.

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